- The market price for the privilege to generate money market returns in the US banking sector relative to foreign alternatives.

(Capital controls are obviously the main driving force of basis swaps between onshore and offshore. These basis swaps are not treated in this book, which assesses the flows between comparatively free markets only.)

Pricing the CCBS is therefore a complex task, which should reflect the complexity of its driving forces. At a minimum any pricing model should include the global asset swap spreads (e.g. of AA-rated corporate issuers) and a proxy for bank risk among its input variables. As proxy for bank risk, the secured–unsecured basis might be used, which offers the advantage of being closely linked to the cost of equity (an input variable into the swap spread model) but observable (e.g. via the price for SOFR–Eurodollar or Fed Funds future contract spreads) and of driving the ICBS as well. Figure 14.4 illustrates the relationship of that basis to the CCBS.

A starting point for modeling the USD–EUR CCBS could therefore be a multiple regression of its price versus the difference of swap spreads of AA-rated corporates in both currencies and versus the SOFR–Fed Funds future spread. Due to the mutual global influences, CCBS for the major currencies should be priced together. This expands the starting point to simultaneously optimizing the USD–EUR, USD–JPY, and USD–GBP CCBS versus the swap spreads of AA-rated corporates in USD, EUR, JPY, and GBP as well as the SOFR–Fed Funds future spreads.

In addition to setting up the pricing model framework, analysts of the CCBS need to be constantly aware of all global developments and their mutual links. In the following we will provide some typical illustrations, starting with the impact of different swap spreads of government bonds on the pricing of the 2Y JPY CCBS in 2002 and 2003. As depicted in Figure 15.1, when the swap spreads of US Treasuries narrowed relative to those of JGBs (i.e. when investing in US bonds became more attractive relative to investing in Japanese bonds), the demand for generating returns in USD increased and the CCBS became more negative. The $R^2$ of 0.34 in the regression of 2Y JPY CCBS on the asset swap spread difference between US Treasuries and JGBs suggests that this effect could be responsible for about one-third of the movements in the CCBS, with the remaining two-thirds being a function of other driving forces, such as the demand for generating money market returns with US banks. This is in line with the CCBS reflecting the result of a multitude of different global funding and investment flows.

To illustrate the impact of a banking crisis on the CCBS, we regress the level of JPY CCBS on a proxy for risk aversion since 2010. In particular, we use the first factor from a PCA of the CCBS curve as the dependent variable. And as the independent variable, we use the realized volatility of the 5Y JPY

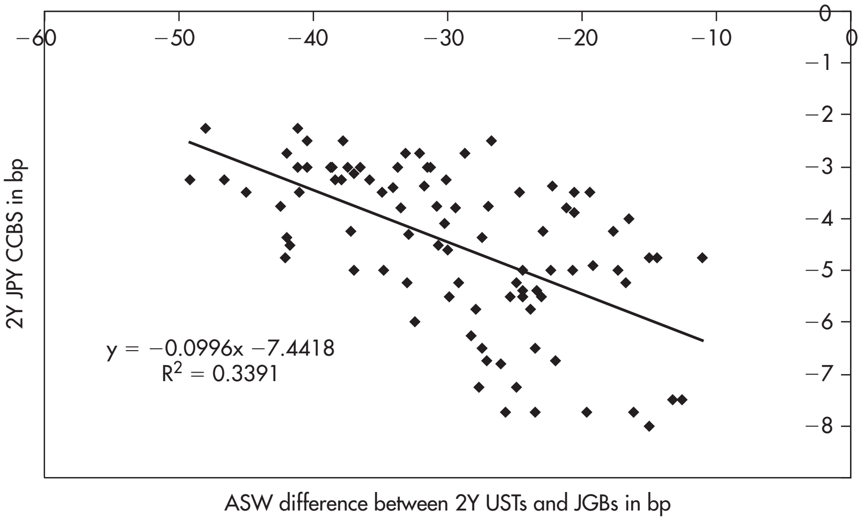


**FIGURE 15.1** The 2Y JPY CCBS in 2002 and 2003 as a function of the ASW difference in different currencies.
*Sources:* data – Bloomberg; chart – Authors.

*Note:* In the 10Y sector, the UST ASW in 2002 and 2003 was largely influenced by (waning) expectations about Fed buying at the long end of the curve. Thus, we use 2Y rather than 10Y (or factor 1 of the PCA on the CCBS curve), where results could be misleading.
*Data period:* 1 Jan 2002 to 7 Oct 2003, weekly data.

swap rate. The results shown in Figure 15.2 underscore the point that in the presence of a banking crisis there is little room for global funding and investment flows to impact the level of the CCBS. Correspondingly, rather than a multitude of driving forces (e.g. Figure 15.1 with an $R^2$ of 0.34), there is now a single determining variable of the CCBS curve only (e.g. Figure 15.2 with an $R^2$ of 0.86). Also note the difference in magnitude. While global funding flows usually cause the CCBS to move by a couple of basis points, a banking crisis can have an impact as much as 10 times larger.

## CCBS and the Subprime Crisis

With the advent of the subprime crisis beginning in 2008, European banks experienced increasing difficulties funding many of their USD-denominated assets, particularly mortgage-related bonds that had been financed in the US repo market.

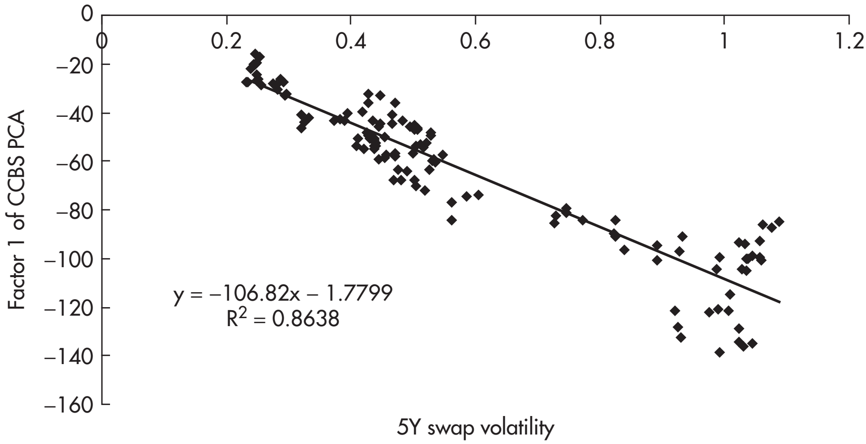


**FIGURE 15.2** Factor 1 of a PCA on the JPY CCBS since 2010 as a function of the 5Y realized swap volatility.
*Sources:* data – Bloomberg; chart – Authors.

*Data period:* 4 Jan 2010 to 4 Jun 2012, weekly data.

One alternative option available to these banks was to sell these mortgage-related assets. But this alternative was considered relatively unattractive, as these assets were selling for relatively low prices.

Another alternative available to European banks was to sell euros to buy dollars in the spot FX market and then to use those dollars to fund their USD mortgage-related assets. But this alternative would involve considerable exchange rate risk.[2]

A third option, and the one chosen by many European banks, was to engage in CCBS with US banks, in which the European bank would:

- receive dollars in exchange for euros at the start of the swap;
- return the EUR and USD principal at the end of the swap;
- exchange ongoing interest payments – plus or minus a basis swap spread – during the life of the swap.

The European banks then used the dollars received in the CCBS to fund their USD mortgage-related investments.

[2]The observant reader might ask whether these European banks could have hedged this FX risk in the forward market. But, of course, combining a spot FX transaction with a forward FX transaction results in an FX swap or in a CCBS (depending on the term of the swap). So from a practical perspective, this is what the European banks did.

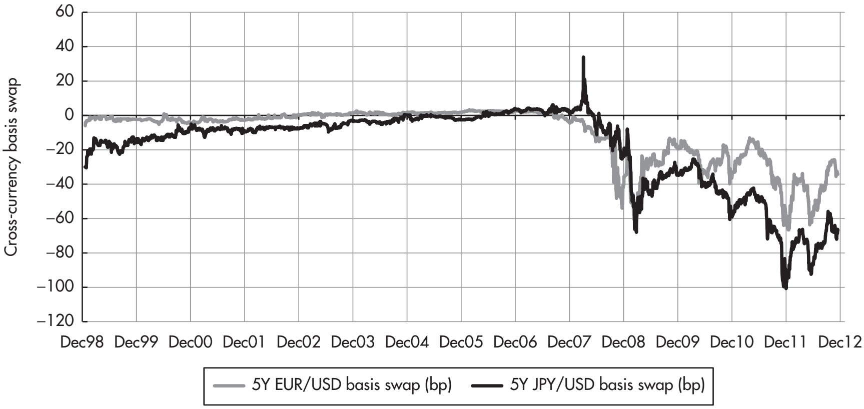


**FIGURE 15.3** 5Y basis swaps: EUR/USD and JPY/USD.
*Sources:* data – Bloomberg; chart – Authors.

Figure 15.3 shows the way in which the 5Y EUR/USD CCBS reacted to these events. The 5Y JPY/USD CCBS is shown as well, for reference.[3]

A couple of points worth noting:

- The basis swap spread was within a few basis points of zero until the start of the subprime crisis, at which time it began increasing (i.e. becoming increasingly negative, favoring the counterparty providing USD in the swap).
- Since the start of the subprime crisis, the 5Y CCBS spread has widened during periods of particular stress for European banks and has narrowed when those stresses have lessened.

## US Dollars 'Going Special' as Collateral

A helpful analogy for the impact of banking crises on the CCBS is given by the perspective of collateralized loans, similar to the repo market for government bonds.

Recall that government bonds sometimes trade in the repo market at rates lower than the "general collateral" (GC) rates, as an incentive for people who are long the bond to provide it to people who are short the bond. The demand for the bond, the specific collateral in the loan, is the factor that drives the repo rate for the bond below the GC repo rate.

[3]In 1998, Japanese banks were in a situation similar to the one described above and used the CCBS to fund their extensive US operations.

Likewise, in an FX swap or a CCBS, we can think of one currency as having particularly strong demand that increases its value as specific collateral. For example, if the 5Y EUR/USD CCBS is -37 bp, the market is willing to provide an incentive of 37 bp per year to motivate people with dollars to provide them as collateral in a loan of euros. In this case, we can think of dollars as having "gone special" as collateral, similar to the way bonds go special in the repo market.

Seen from this perspective, the arbitrage that is available for providing dollars in exchange for euros for a fixed period is the incentive provided to people who are long dollars to lend them to people who are short dollars, against a loan of euros.

In contrast, consider the situation in which a comparable loan of euros is securitized not with US dollars but rather with some more GC, say, gold or generic German Bunds. In this case, the interest rate paid by the person borrowing euros would be the generic rate for loans secured by GC (i.e. the GC repo rate). But when the euro loan is secured by US dollars, which are in short supply among European banks, the party borrowing euros enjoys a special borrowing rate as an incentive for providing the dollars for a fixed period.

When a bond becomes special in the repo market, it assumes a value to its owner beyond the mere right to the intrinsic payments of principal and interest. The bond also allows its owner to borrow in the secured lending market at an interest rate lower than he could borrow with GC. In fact, by combining the resulting low-rate borrowing with collateralized lending at the higher rate prevailing for GC, the owner of the special collateral can monetize the specialness of his valuable collateral without incurring interest rate risk.

Since the bond trading special in the repo market offers its owner this additional economic value, its price in the spot market tends to be higher than it would be were it not special in the repo market. In other words, the spot price of a bond tends to increase relative to other bonds when it goes special in the repo market.

This perspective that CCBS spreads widen when dollars 'go special' in the repo market implies that US dollars should richen to euros in the spot market above the rate they otherwise would be were the dollars not special (i.e. were the basis swap not so wide).

Of course, the CCBS is by no means the only factor that would be expected to affect the price of US dollars vs. euros in the spot FX market, so a failure to observe the predicted relationship may not warrant a rejection of our hypothesized perspective. On the other hand, if we were to observe the hypothesized relationship, it should lend credence to our analogy between repo specialness and CCBS spreads.

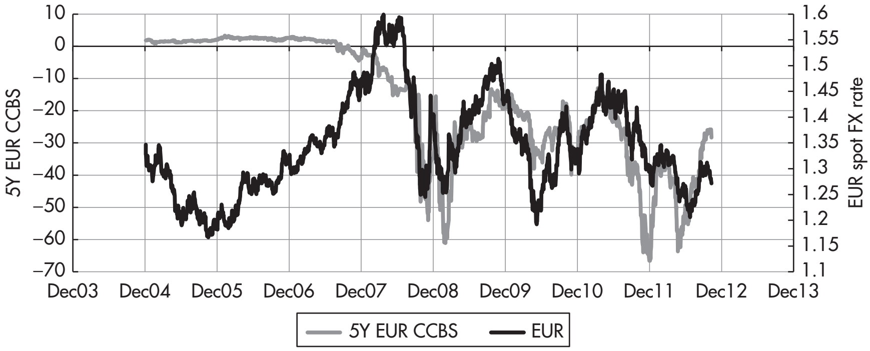


**FIGURE 15.4** 5Y EUR/USD CCBS versus the EUR/USD spot FX rate.
*Sources:* data – Bloomberg; chart – Authors.

Figure 15.4 shows the relation between the 5Y EUR/USD CCBS and the EUR/USD spot exchange rate over time since January of 2005. Prior to the start of the subprime crisis, there was no correlation between the basis swap spread and the spot value of the EUR, particularly as the value of the basis swap spread always was within a few basis points of zero.

After the start of the subprime crisis, the correlation between the CCBS spread and the FX rate increased significantly. For example, in 2009, 2010, and 2011 the correlations between the two series were 0.84, 0.80, and 0.78 respectively. In our view, the relation between the CCBS spread and the value of the dollar relative to the euro is consistent with our perspective that the CCBS spread can be viewed as the extent to which one currency has become more valuable as collateral against a loan in the other currency.

As the dollar tends to richen in the spot market, the other currency tends to appear relatively rich in the forward FX market. Correspondingly, the deviation of the forward FX rate traded in the market from the arbitrage relationship described by the equation above is often referred to as the 'richness' of the other currency (e.g. EUR) in the forward FX market.

With this in mind, our view is that a main economic function of the FX swap and CCBS markets is to provide an opportunity for people to borrow in one currency using another currency as collateral. As a result, many of the results we have developed for collateralized lending can be applied in this setting as well, the only complication being that the currency in which the collateral is denominated in this case is other than the currency in which the borrowed funds are denominated.

## THE IMPACT OF THE TRANSITION TO NEW REFERENCE RATES ON THE CCBS

The transition away from LIBOR to other reference rates explained in Chapter 11 had several impacts on the definition, conventions, and pricing of the CCBS:

- The move from the term rate LIBOR to O/N rates had the economic impact of reducing the term risk in the relevant leg of the CCBS. Consider again the behavior of the USD–EUR CCBS in times of a banking crisis: even when both legs were LIBOR, the preference to keep money in the US banking system rather than abroad during times of stress caused the CCBS to quote more negative in these situations. Now that the US reference rate is a secured O/N rate, i.e. the safest reference rate worldwide, the USD SOFR–EUR 3M EURIBOR CCBS can be expected to react even more to banking crises than the former USD 3M LIBOR–EUR 3M EURIBOR CCBS.
- The move from the term rate LIBOR to O/N rates had the technical consequence of determining the payments by compounding in arrears rather than via the term rate LIBOR known in advance of each period. For example, in the formula for the FX swap above, when $R_{USD}$ represented LIBOR, it was observable at the beginning of the FX swap; now that it represents SOFR, it needs to be determined by aggregating all SOFR values observed during the life of the FX swap, hence in arrears. Given the resulting practical problems, several mitigating features have been introduced, including a SOFR term rate. These are described extensively in Chapter 3 of Huggins and Schaller (2022).
- The move from the unsecured rate LIBOR to the secured rate SOFR means that the secured–unsecured basis is part of any CCBS currently trading and hence that the pricing model for this spread outlined in Chapter 11 is a natural part of the pricing model of any CCBS. While in the model for CCBS with LIBOR legs sketched above, the secured–unsecured basis was only one possible proxy for stress in the banking sector, for CCBS with a USD SOFR leg, it is also an integral part of the CCBS and its prices. This further underlines the advantage of using the secured–unsecured basis as input variable for CCBS pricing models.

Before the transition away from LIBOR, CCBS exchanged unsecured term rates. Even then, the preference to keep funds in the US banking sector in times of crises has led to a clear correlation of the CCBS (more negative) to variables measuring stress, such as the ICBS, volatility, cost of capital, etc. Following the transition to SOFR in the US, CCBS now always exchange the unsecured foreign rate versus the secured O/N rate SOFR in the US. As the secured–unsecured basis is also related to stress (given its dependency on the cost of capital), the CCBS are now exposed twofold to this basis, as before due to the preference to keep money in the US banking system and additionally due to the preference to lend money secured.

On the foreign leg there exist only unsecured reference rates, in some countries with different terms. In case of the foreign leg being LIBOR and thus including term risk, the difference in credit risk covered by the CCBS is even larger than in case of the foreign leg being an O/N rate. From the starting point of CCBS using LIBOR on both legs, one can therefore make the qualitative statements:

- CCBS exchanging USD SOFR versus a foreign unsecured O/N rate are generally more negative than the CCBS before the transition, *ceteris paribus*, since they directly include the secured–unsecured basis in addition to the swap between two banking systems (which is only indirectly correlated to the secured–unsecured basis as a proxy for stress in the banking sector).
- CCBS exchanging USD SOFR versus a foreign unsecured term rate (LIBOR) are expected to be even more negative, since they also include term risk. For example, the SOFR–ESTR CCBS may quote at −20 bp, while the SOFR–EURIBOR CCBS may quote at −28 bp.
- Hence, the variable secured–unsecured basis drives both the general level of all types of CCBS (since all of them reflect the preference for keeping funds in the US banking system) and the spread between CCBS with secured SOFR as one leg and unsecured O/N rates as other leg. Furthermore, as it is correlated to the term (risk) (the more demand for secured versus unsecured lending, the more demand for shorter versus longer unsecured lending), it also drives the spread to CCBS with LIBOR as the non-USD leg. Altogether, this variable explains the observation that a general increase in the level of all CCBS tends to be accompanied by increasing spreads between the different types.
- The transition has therefore resulted in an even stronger impact of the secured–unsecured basis (and thus the banking crises reflected in it) on the CCBS. While already the CCBS with LIBOR in both legs were significantly driven by that variable (see Figure 14.4), the CCBS between USD SOFR and non-USD LIBOR are generally more negative and more volatile.

Quantifying these qualitative statements, we can price all types of CCBS by adding to the CCBS between LIBOR the models for spreads between reference rates in the same currency. From the starting point of the CCBS between LIBOR on both legs described above, the current types of CCBS can be reconstructed by combining them with an ICBS between LIBOR and SOFR in the

USD and depending on the type also with an ICBS between LIBOR and an O/N reference rate in the foreign currency. For example, a SOFR–EURIBOR CCBS can be priced by adding the fair value of a SOFR–USD LIBOR ICBS to the CCBS with LIBOR on both legs. As long as the SOFR–Eurodollar spread contracts trade, there is even a market price for the SOFR–USD LIBOR ICBS: after the end of Eurodollar futures, our model for the repo–LIBOR spread can still provide the input variable. If for the CCBS pricing model sketched above the secured–unsecured basis is chosen as proxy for stress in the banking system, it can also be used as input variable for the ICBS pricing model (and for the asset swap spread model added to the combinations in Chapter 16).

In terms of Table 11.1, the CCBS with LIBOR on both legs exchange different rows (currencies) of the left-most column. By combining them with ICBS, any type of CCBS can be replicated. Hence, any type of CCBS can be priced by adding to the pricing model for CCBS with LIBOR the fair value of the ICBS obtained in Chapter 14 via a direct application of the models for the spreads between reference rates.

There are several ways of reaching the same goal: as an alternative to the "historical" start above from CCBS with LIBOR on both legs, one could also start with a hypothetical CCBS with repo on both legs and the add the fair value of the repo–LIBOR spread to price USD SOFR–EUR EURIBOR CCBS.

It is important to note that once more the transition has introduced a structural break in the time series of CCBS: before the transition, CCBS data reflect LIBOR in both legs, after the transition, CCBS data reflect SOFR in one leg, hence a structurally higher and more volatile time series for the reasons mentioned above. Analysts should therefore take care to check the type of "CCBS" represented by the time series they use. It may be worthwhile constructing synthetic time series of CCBS with LIBOR in both legs (or with repo in both legs, or with unsecured O/N rates in both legs) via the methods just explained, i.e. by adjusting other types of CCBS quotes with the (fair value of the) ICBS.

As a practical consequence for this book written in 2023, given the short time since the transition, there are too few data for CCBS with SOFR as one leg in order to generate long-term time series without structural breaks. We have therefore kept most of the case studies from the first edition written in 2012 before the transition, i.e. with LIBOR in both legs of the CCBS. While this decision requires carrying around some historical ballast, it avoids contaminating the case studies with structural breaks. And that point in time provided with the subprime and euro crisis interesting precedents, which remain worthwhile to study. Of course, once enough CCBS data post transition becomes available in future, the reader is encouraged to re-apply the analysis techniques described here.

# CHAPTER 16

# Combinations and Mutual Influences of Asset, Basis, and Credit Default Swaps

## INTRODUCTION

Having discussed the different types of swaps individually in the previous chapters, the objective of this chapter is the analysis of their combinations and the resulting mutual influences between all asset, basis, and credit default swaps worldwide.

The first section explains the basics of calculating USD asset swap spreads of bonds denominated in other currencies, i.e. their spread versus USD SOFR (or formerly, versus USD LIBOR) as determined by asset and basis swapping them. The details of executing these operations are given in Chapter 17.

This ability to express all bonds as a spread versus USD SOFR (formerly LIBOR), regardless of their issuer, currency, and maturity, serves as a common yardstick, against which all bonds can be compared. The existence of such a possibility to price all bonds versus one single reference rate has major consequences:

- It is the foundation for the funding and investment arbitrage between different markets described in Chapter 15.
- It provides a natural and universal rich/cheap measure for bonds, which enables RV assessments between global bonds. This is the topic of Chapter 17.
- It links all global bond markets via the asset and basis swap markets. These links and the resulting mutual influences and relationships are the topic of this chapter.

Furthermore, just as the operations of asset and basis swapping a bond express its pricing in a single number, i.e. its spread versus USD SOFR, so does the credit default swap (CDS). There exist therefore two different ways of measuring every bond worldwide as a spread over USD SOFR: The asset swap spread (of the basis swapped bond) and the CDS (of the bond issuer for